Project Design Phase-II

Data Flow Diagram and User Stories

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Project Title: FORMULA 1 Race Intelligence Dashboard Analysis Report (2010 – 2026)		**Domain:** Data Analytics with Tableau

1. Data Flow Diagram (DFD Architecture)

Level 0 & 1 Data Pipeline:
[Raw Excel Sources: circuits, championship_standings, lap_intelligence, qualifying_results, race_results]
➔ [ETL & Cleansing: Null Handling, Datetime parsing, Compound standardization]
➔ [Data Model: Star Schema centered on race_results (6,464 records)]
➔ [Tableau Analytical Engine: Calculated Fields, LOD Expressions, Parameter Filters]
➔ [Output Layer: Strategic Decision Simulator, Broadcast Media Visuals, FIA Compliance Reports]

2. Agile User Stories

Story ID	As a (Role)	I want to...	So that...	Acceptance Criteria
US-01	Team Strategist	Compare my constructor's points trajectory with Red Bull & Mercedes	I can measure our championship points deficit and set seasonal performance targets	Multi-line chart displaying 2010–2026 points with team color filters
US-02	Team Strategist	Analyze tyre compound pace deltas across permanent and street circuit types	I can determine whether a 1-stop or 2-stop pit strategy offsets the 22.4s pit loss	Heatmap displaying Soft, Med, Hard lap deltas across circuit categories
US-03	Broadcast Analyst	Filter cumulative driver points, wins, and podiums across all 17 seasons	I can generate accurate on-screen TV graphics for pre-race commentary	Sorted horizontal bar chart displaying top 10 drivers with dynamic metric switch
US-04	FIA Analyst	Track Did Not Finish (DNF) causes across regulatory engine eras	I can audit if engine regulations reduced powertrain blowouts vs collisions	100% stacked bar chart categorizing Engine, Collision, Gearbox, and Brakes